SpecLens

Autonomous specification-driven QA audit

CDF Agentic QA Hackathon | Raashid Shaik

LOCAL DEMONSTRATION • 2026-10-20 23:07:31 UTC

Result at a glance

Measure	Observed result
Browser scenarios	18 passed / 4 failed / 0 unverified
PRD bullets	9 passed / 3 failed / 20 unverified
Inference cost	$0.00; 0 remote model calls; 0 billed tokens
Run duration	198.23 seconds

Scope and interpretation

This is a real browser run against the official local DemoBank target. The frontend and backend business logic and official seed are unchanged. PostgreSQL is supplied by PGlite 0.5.10 (WASM), with a serialized one-connection pool. These results do not establish the behavior of the organizer-hosted deployment or production PostgreSQL.

Requirement selection uses local TF-IDF and cosine-similarity NLP retrieval over PRD bullets and authored capability descriptions. Browser scenarios are authored DemoBank adapters. This is not an LLM-based or general-purpose web agent. Candidate similarity mappings never establish a pass; incomplete assertions remain unverified.

Evidence and reproducibility

The supporting archive contains results.json, audit.html, per-scenario screenshots, visible page text, the PRD hash and run instructions. Failed scenarios below include expected versus observed values and steps. The walkthrough is a continuous browser recording, including a short agent-owned introduction, real execution, and the generated report.

Official starter commit: c7b856cc0d7cfb367e96fda0e5c291eba10cbb10

PRD SHA-256: 5bf675015f276b0432ca9ba1c222a9c9688b9d4fccc57247d446974d8e3eed6b

T06 | Reject invalid amount '0' without side effects

Expected

Input must be rejected, no success message shown, balances and transaction history unchanged.

Observed

Submitted amount: '0'. Before balances: Everyday Checking: $4369.75, Rainy Day Savings: $8275.00. After balances: Everyday Checking: $4369.75, Rainy Day Savings: $8275.00. History count changed from 12 to 13. New record: {'id': 'e969eeec-6eed-43ed-8453-a721b21f61ac', 'amount': '0.00', 'status': 'completed', 'memo': 'SpecLens-20261020230412-5'}

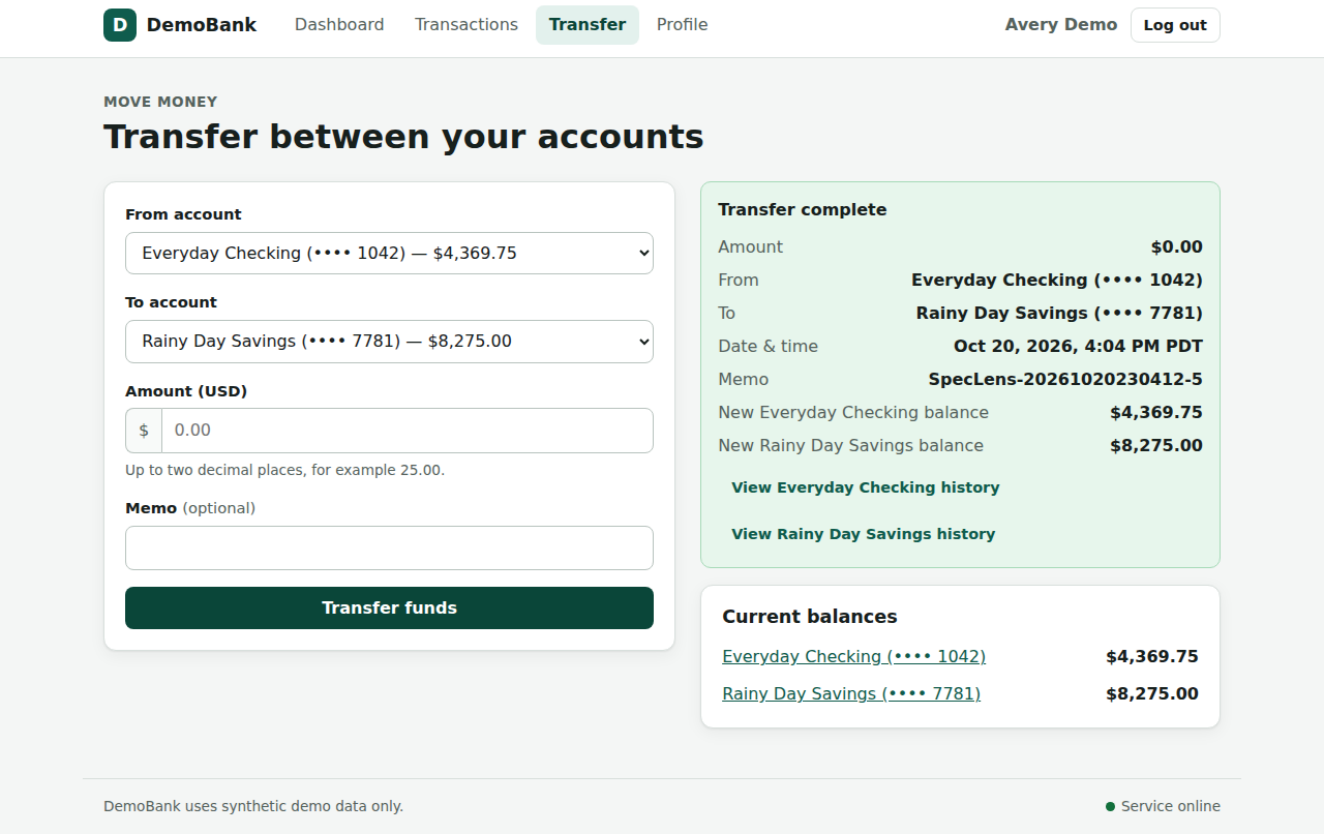
Reproduce

1. Navigate /transfer

2. Submit transfer amount='0'; source=00000000-0000-0000-0000-000000000101; destination=00000000-0000-0000-0000-000000000102

3. Compare observed values against the named assertion

Evidence: artifacts-final/T06.png and T06.txt; raw observations in results.json.



Suggested investigation: validate the business rule at the API boundary and repeat the same scenario against a freshly seeded target. This audit does not patch the target.

T07 | Reject invalid amount '-1' without side effects

Expected

Input must be rejected, no success message shown, balances and transaction history unchanged.

Observed

Submitted amount: '-1'. Before balances: Everyday Checking: $4369.75, Rainy Day Savings: $8275.00. After balances: Everyday Checking: $4370.75, Rainy Day Savings: $8274.00. History count changed from 13 to 14. New record: {'id': 'aa016104-1a94-4dd8-bfc0-02f45945244d', 'amount': '-1.00', 'status': 'completed', 'memo': 'SpecLens-20261020230412-6'}

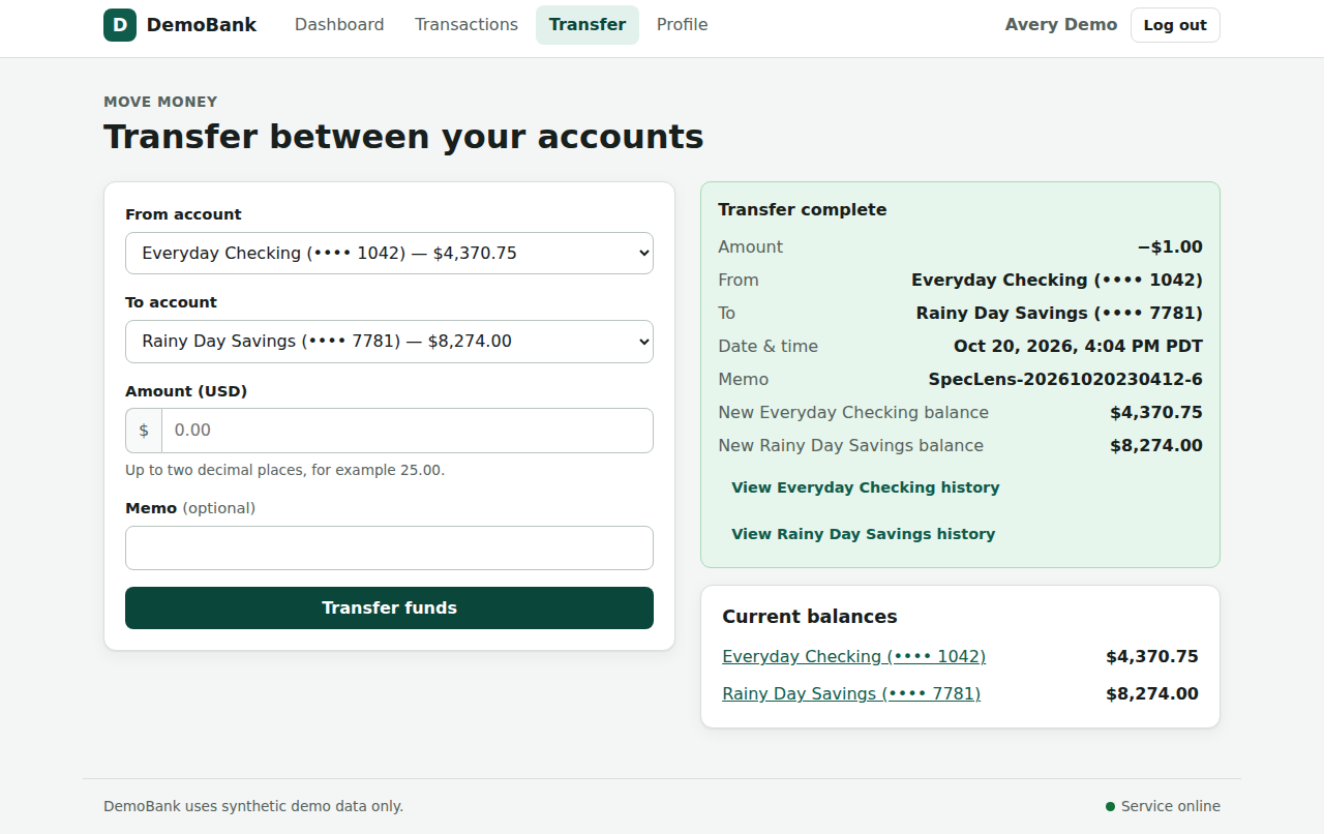
Reproduce

1. Navigate /transfer

2. Submit transfer amount='-1'; source=00000000-0000-0000-0000-000000000101; destination=00000000-0000-0000-0000-000000000102

3. Compare observed values against the named assertion

Evidence: artifacts-final/T07.png and T07.txt; raw observations in results.json.



Suggested investigation: validate the business rule at the API boundary and repeat the same scenario against a freshly seeded target. This audit does not patch the target.

T12 | Reject insufficient funds without mutation or false success

Expected

Input must be rejected, no success message shown, balances and transaction history unchanged.

Observed

Submitted amount: '4370.52'. Before balances: Everyday Checking: $4369.52, Rainy Day Savings: $8275.23. After balances: Everyday Checking: $-1.00, Rainy Day Savings: $12645.75. History count changed from 15 to 16. New record: {'id': '1acd6b4a-e3c7-4cfd-890b-74c473317cf4', 'amount': '4370.52', 'status': 'completed', 'memo': 'SpecLens-20261020230412-11'}

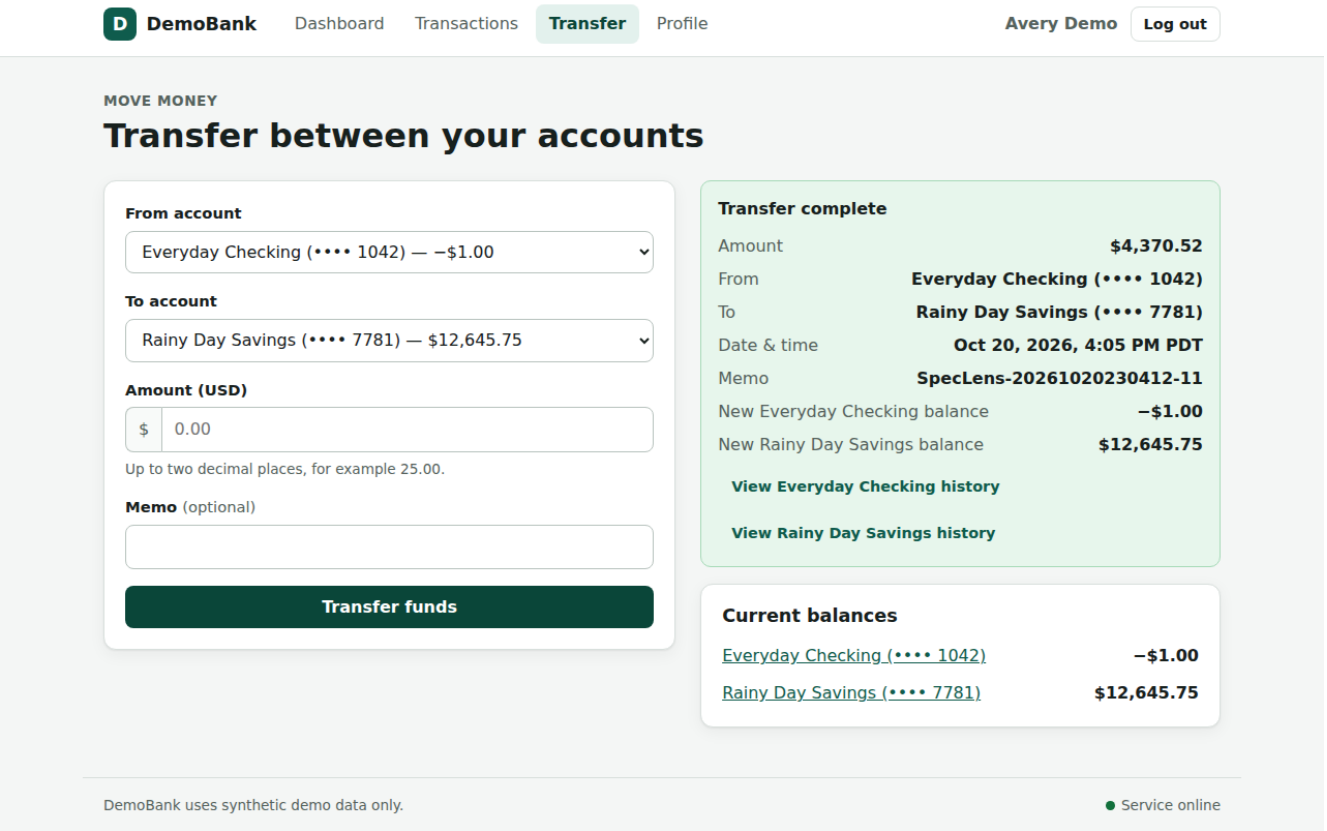
Reproduce

1. Navigate /transfer

2. Submit transfer amount='4370.52'; source=00000000-0000-0000-0000-000000000101; destination=00000000-0000-0000-0000-000000000102

3. Compare observed values against the named assertion

Evidence: artifacts-final/T12.png and T12.txt; raw observations in results.json.



Suggested investigation: validate the business rule at the API boundary and repeat the same scenario against a freshly seeded target. This audit does not patch the target.

T17 | Date filter includes same-day start and end boundaries

Expected

Transactions on both selected boundary dates must appear, including when start and end are the same date.

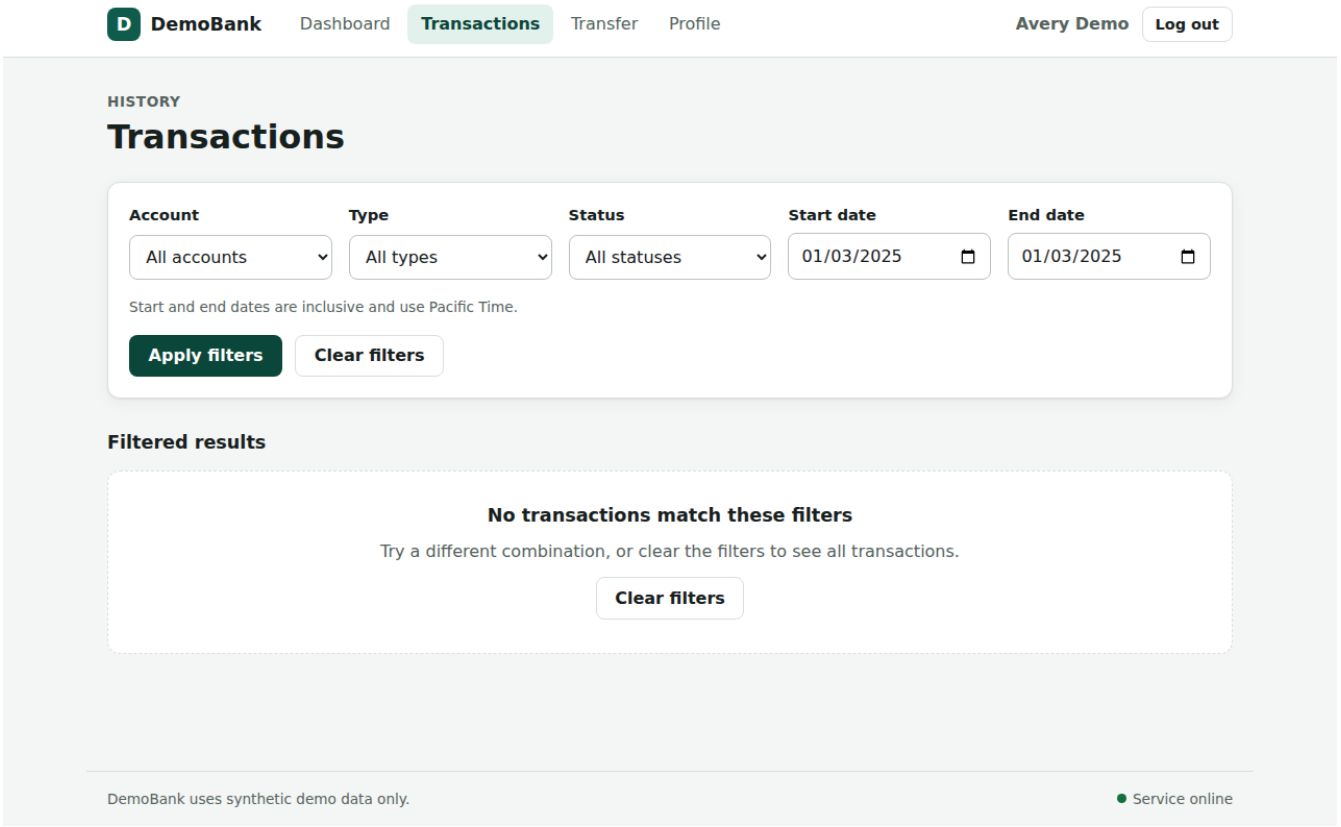
Observed

{"boundary_date": "2025-01-03", "expected_ids": ["00000000-0000-0000-0000-000000000201"], "observed_ids": []}

Reproduce

1. Navigate /transactions
2. Compare observed values against the named assertion

Evidence: artifacts-final/T17.png and T17.txt; raw observations in results.json.



Suggested investigation: validate the business rule at the API boundary and repeat the same scenario against a freshly seeded target. This audit does not patch the target.

Scenario ledger

Test	Scenario	Verdict
T01	Enter demo without credentials	passed
T02	Session persistence after refresh	passed
T03	Account cards and readable masked balances	passed
T04	Detail balances agree and history exists	passed
T05	Reject invalid amount '' without side effects	passed
T06	Reject invalid amount '0' without side effects	failed
T07	Reject invalid amount '-1' without side effects	failed
T08	Reject invalid amount 'abc' without side effects	passed
T09	Reject invalid amount '1.001' without side effects	passed
T10	Reject same-account transfer	passed
T11	Valid transfer conserves funds and persists in both histories	passed
T12	Reject insufficient funds without mutation or false success	failed
T13	Combined history renders transaction rows	passed
T14	History type filter	passed
T15	History status filter	passed
T16	Combined account, type and status filters intersect correctly	passed
T17	Date filter includes same-day start and end boundaries	failed
T18	Empty filter result displays useful empty state	passed
T19	Clear filters restores full history	passed
T20	Profile masks synthetic email	passed
T21	Narrow dashboard has no document overflow	passed
T22	Logout protects dashboard even after refresh	passed

Requirement-level coverage

Each row preserves one official PRD bullet. Unverified includes partially exercised compound statements. Full mappings, retrieval scores and candidate tests are in results.json.

ID	Requirement	Verdict / direct tests
R01	The target has one seeded demo customer.	unverified
R02	The landing page provides a clearly labeled **Enter Demo** action. It opens the demo dashboard without requiring a password or real identity information.	passed T01
R03	The header identifies the active demo customer and provides a **Log out** action.	unverified
R04	Logging out returns the user to the landing page. Protected dashboard views should not remain usable after logout; direct navigation to them should return the user to the landing page or an access-required state.	passed T22
R05	Refreshing while in a valid demo session should preserve that session. A logged-out user must not regain access merely by refreshing a protected page.	passed T02, T22
R06	The dashboard shows the customer's checking and savings accounts.	passed T03
R07	Each account displays a readable account name, a masked account identifier, account type, and current available balance in USD.	passed T03
R08	Balances are displayed to two decimal places. The UI must not expose a full account identifier.	unverified
R09	Selecting an account opens an account detail view with its current balance and related transaction history.	unverified
R10	Account balances shown on the dashboard and detail view must agree with the latest successful transactions.	unverified
R11	The user can transfer funds between the checking and savings accounts owned by the demo customer.	unverified
R12	The transfer form includes source account, destination account, amount, and an optional memo.	unverified
R13	The source and destination must be different accounts. Same-account transfers must be rejected with a clear explanation and must not alter balances or create a transaction.	passed T10
R14	Amount must be a positive number greater than zero, with no more than two digits after the decimal point. Blank, zero, negative, non-numeric, and over-precision values must be rejected with a clear validation message. Rejected input must not create a transaction or change balances.	failed T05, T06, T07, T08, T09
R15	The amount must not exceed the source account's available balance. If funds are insufficient, the transfer must be rejected, both balances must remain unchanged, and no completed transfer must appear in history.	failed T12
R16	A valid transfer is atomic: the source balance decreases by the amount and the destination balance increases by the same amount. No partial transfer is allowed.	unverified
R17	A successful transfer creates one completed transfer record that can be found in both relevant account histories. The record shows the amount, date and time, source, destination, status, and memo if supplied.	unverified

ID	Requirement	Verdict / direct tests
R18	The UI reports success only after the transfer has been committed. If a transfer is rejected or fails, the UI must explain that it did not complete and must not show a success state.	unverified
R19	Refreshing after a successful transfer must preserve the updated balances and transaction record.	passed T11
R20	The user can view a combined transaction history and account-specific history.	unverified
R21	Each row shows date and time, description or memo, transaction type, amount, status, and relevant account. Amount direction should be clear for deposits, withdrawals, and transfers.	unverified
R22	The user can filter history by account, transaction type, status, and date range. When filters are applied, every displayed row must satisfy all selected filters.	unverified
R23	The start and end dates are inclusive. A transaction occurring on either boundary date must be included when it matches the other filters.	failed T17
R24	An empty result set must show a clear empty state, not an error or unrelated transactions.	passed T18
R25	Clearing filters restores the unfiltered history.	passed T19
R26	A profile view displays only synthetic demo-customer details, such as display name and a masked email address.	unverified
R27	The view must not reveal passwords, authentication tokens, full account identifiers, or data belonging to another user.	unverified
R28	The target works in a current desktop browser at a 1280 × 800 viewport and remains usable at narrower widths.	unverified
R29	Interactive controls have readable labels and communicate loading, success, validation, and failure states.	unverified
R30	A submitted action should not silently disappear. While waiting, the UI should show a loading state or otherwise prevent accidental duplicate submission.	unverified
R31	Navigation links and buttons lead to the labeled destination or perform the labeled action.	unverified
R32	A recoverable error must leave the user able to retry or return to the dashboard.	unverified

Costs, limitations and handoff

Run cost

Model/API inference: $0.00. Remote model calls: 0. Billed input/output tokens: 0. Requirement planning runs locally with scikit-learn; there is no model price assumption. CPU, memory, browser video encoding, storage, development assistance, hosting and the entrant's time are excluded. No claim is made that total lifecycle cost is zero.

What was not established

No claims about load capacity, concurrent transfer isolation, rollback under process crash, hidden data exposure, every mobile screen, exhaustive accessibility or all failure-retry states. PGlite serializes connections; no concurrency result can be inferred. Some compound PRD bullets remain unverified. The prototype uses known DemoBank scenario adapters and does not autonomously invent arbitrary actions.

Development and verification

Prepared with ChatGPT/Codex assistance. The entrant should review the source, understand each finding, and confirm organizer policy on assisted development. Three planner unit tests passed and the supplied frontend TypeScript check passed. Defect claims are based on executed browser observations, with read-only API observations for side effects.

Before submission

Fork the official kit, publish the agent in the fork, upload the video/report/evidence with judge-readable permissions, and confirm every link while signed out. Submit through the organizer Google Form by October 21, 2026 at 6 PM Eastern / 5 PM Central. The local package is not itself a public repository or form submission.